

§1 — Compact Internal Sector (A1 Primary Route)

We begin with definitions only. No promises.

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1.1 Internal Sector Definition

Define the internal regulator manifold:

$$\begin{aligned} &\mathcal{I}_3 \\ &= \\ &S^1_{\zeta} \\ &\times \\ &S^1_{\omega} \\ &\times \\ &S^1_{\xi} \end{aligned}$$

with coordinates

$$\begin{aligned} y_{\zeta} &\sim y_{\zeta} + L_{\zeta}, \\ \quad &\quad \\ y_{\omega} &\sim y_{\omega} + L_{\omega}, \\ \quad &\quad \\ y_{\xi} &\sim y_{\xi} + L_{\xi}. \end{aligned}$$

The coordinate periods L_a are finite and strictly positive.

No empirical fixing is assumed at this stage.

1.2 Internal Metric Specification

Adopt a diagonal regulator-consistent ansatz:

$$\begin{aligned} &h_{ab} \\ &= \\ &\mathrm{diag} \\ &\big(\\ &\zeta^2, \\ &\omega^2, \\ &\xi^2 \end{aligned}$$

$\big)$.

At this stage we classify:

- Option A: ζ, ω, ξ are constant moduli (static compactification).
- Option B: They are slowly varying moduli fields over external spacetime.

The induced measure is

$$\sqrt{\det h}$$

$$=$$

$$\zeta \, \omega \, \xi.$$

1.3 Computation of λ

By definition,

$$\lambda$$

$$=$$

$$\int_{\mathcal{I}_3} \sqrt{\det h} \, d^3 y.$$

For the compact product space:

$$\lambda$$

$$=$$

$$\int_0^{L_\zeta} \int_0^{L_\omega} \int_0^{L_\xi} \zeta \, \omega \, \xi \, dy_\zeta dy_\omega dy_\xi.$$

Since the measure factor is independent of internal coordinates under Option A,

$$\boxed{\lambda}$$

$$=$$

$$(\zeta \, \omega \, \xi)$$

$$\int_0^{L_\zeta} \int_0^{L_\omega} \int_0^{L_\xi} dy_\zeta dy_\omega dy_\xi.$$

This is finite if and only if all L_a are finite.

1.4 Period Classification Fork

We now classify the status of the periods L_a .

Case A — Physics-Fixed Periods

If:

- L_a are determined by topology, extension bounds (Appx_10), or regulator irreducibility constraints,

then λ is fully determined by upstream structure.

Prediction power preserved.

Case B — Gauge/Convention Periods

If:

- Rescaling L_a can be absorbed into coordinate redefinition without affecting observables,

then λ may be conventional and only the ratio κ_7/λ matters.

This must be resolved by §4 consistency lock.

1.5 Local Failure Modes (A1)

A1 fails if:

- Any L_a must be chosen ad hoc (e.g. “ 2π because convenient”)
- ζ, ω, ξ evolve without satisfying dynamics constraint
- Changing L_a alters α^G while leaving upstream invariants fixed

If any condition above holds, A1 cannot close λ non-empirically.

§2 — Dynamics Constraint (Derived Consequence)

Purpose

Ensure no $\dot{G}/G$ violation emerges from the internal measure behavior (via λ) under the reduction identity. This is consistency policing, not parameter fitting.

2.1 Time-Variation Relation (Identity)

From §4.1:

$$G_4 = \frac{\kappa_7^8 \pi \lambda}{G_4}.$$

Take a time derivative (log form):

$$\boxed{\frac{\dot{G}_4}{G_4} = \frac{\dot{\kappa}_7}{\kappa_7} - \frac{\dot{\lambda}}{\lambda}}$$

This is purely algebraic. Any “gravity drift” comes from drift in κ_7 and/or λ .

2.2 Static-Moduli Case (Preferred)

Assume the reduction data are time-independent over the regime in which GR recovery is asserted:

- $\dot{\kappa}_7 = 0$ (constant normalization / fixed invariant), and
- $\dot{\lambda} = 0$ (static internal measure).

Then:

$$\boxed{\frac{\dot{G}_4}{G_4} = 0}$$

Status: clean closure.

Note: if you adopt this, it must be explicit: “internal periods/moduli are fixed (or effectively frozen) in the recovery regime.” No hand-waving.

2.3 Dynamic-Moduli Case (Allowed, but fenced)

If internal moduli or warp factors vary, then $\lambda = \lambda(t)$ and possibly $\kappa_7 = \kappa_7(t)$.

Two admissible subcases:

(A) κ_7 constant by convention/invariant, $\lambda(t)$ varies
 $\frac{\dot{G}_4}{G_4} = -\frac{\dot{\lambda}}{\lambda}$.

(B) both vary
 $\frac{\dot{G}_4}{G_4} = \frac{\dot{\kappa}_7}{\kappa_7} - \frac{\dot{\lambda}}{\lambda}$.

In either subcase, the framework must provide a mechanism that bounds the RHS from upstream structure, not by picking functions to satisfy data.

If you cannot derive a bound, this route stays TRAILHEAD at best.

2.4 Empirical Bound (Consistency Check Only)

Introduce an external constraint as a post-derivation veto, not a knob:

$\left| \frac{\dot{G}_4}{G_4} \right| \leq \epsilon_{\text{LLR}}$
 $\quad \text{(lunar laser ranging bound, or equivalent)}$.

We do not choose $\lambda(t)$ to satisfy this. We check whether the derived $\lambda(t)$ (or derived moduli dynamics) implies a violation.

This is a kill condition, not a calibration condition.

2.5 Kill-Switch Condition (Quarantine-First Trigger)

Define the operational kill switch:

KILL SWITCH ID: DGDT_BOUND_VIOLATION

Trigger condition:

```
\boxed{
\left|\frac{\dot{\kappa}_7}{\kappa_7}-\frac{\dot{\lambda}}{\lambda}\right|
>
\epsilon_{\text{LLR}}
}
```

Effect (quarantine-first):

- Node becomes FROZEN pending adjudication (since the failure may be due to unproven dynamics assumptions, not necessarily the whole reduction identity).
- If adjudication confirms the framework requires violation (not a missing lemma), promote to FALSIFIED for this channel.

Note: If §1 proves λ is strictly time-independent by construction (compact internal sector + fixed periods), this kill switch becomes inert and can be recorded as “non-applicable under static closure.”

§3 — Non-Compact Fallback (A2) — Write only if §1 fails

Purpose

Provide a protocol-clean alternative proof of $\lambda < \infty$ when the internal sector is non-compact, without reintroducing “pick a warp factor until it converges” tuning.

3.1 Internal Sector Definition (Non-Compact)

Define the internal sector:

- Manifold: $\mathcal{I}_3 \cong \mathbb{R}^3$ (or specified half-lines if needed)
- Coordinates: $y=(y_\zeta,y_\omega,y_\xi)\in \mathbb{R}^3$
- Metric: $h_{ab}(y)$ on $\mathcal{I}_3$
- Reduction measure:

$$\lambda := \int_{\mathcal{I}_3} \sqrt{\det h(y)} \, d^3y$$

Hard requirement: $\sqrt{\det h(y)}$ must be determined from upstream constraints (Appx_10 bounds / Appx_18 metric structure), not selected ad hoc.

3.2 Functional Class Specification (No Free Warp Toys)

Parameterize admissible metrics as:

$$\begin{aligned} h_{\{ab\}}(y) &= \mathrm{diag} \{ \big(f_{\zeta}(y)^2, f_{\omega}(y)^2, f_{\xi}(y)^2 \big) \\ &\quad \rightarrow \\ \sqrt{\det h} &= f_{\zeta} f_{\omega} f_{\xi} \end{aligned}$$

Admissible class $\mathcal{F}$: functions $f_a(y)$ must be:

1. Nonnegative and measurable
2. Asymptotically decaying in $|y|$ at a rate fixed by upstream invariants
3. Non-tunable: no free decay length parameters unless derived

Two acceptable “families” (pick one only if derivable upstream):

- Exponential localization family
 $f_{\omega}(\xi) \leq C e^{-q|\xi|}$
- Power-law localization family
 $f_{\omega}(\xi) \leq C (1+|\xi|)^{-p}$

Where C,q,p are not fit parameters; they must be computed or bounded from Appx_10/18 constraints.

3.3 Integrability Theorem (Finiteness)

Theorem A2.1 (Exponential)

If there exist $C > 0$, $q > 0$ such that for $|y|$ sufficiently large:

$$0 \leq \sqrt{\det h(y)} \leq C e^{-q|y|}$$

then

$$\lambda = \int_{\mathbb{R}^3} \sqrt{\det h(y)} dy < \infty.$$

Theorem A2.2 (Power-law)

If there exist $C > 0$, $p > 3$ such that for $|y|$ sufficiently large:

$$0 \leq \sqrt{\det h(y)} \leq \frac{C}{(1+|y|)^p}$$

then

$$\lambda < \infty.$$

Proof sketch: standard comparison test against known integrable radial envelopes in $\mathbb{R}^3$.

3.4 Derived Decay Scales From Regulator Bounds (Non-Empirical)

This is the critical closure point: define decay rates from upstream constraints.

Template rule (must be instantiated from Appx_10/18):

- Let $\mathcal{B}$ be the bound-set from Appx_10 (extension/constraint admissibility)
- Let $\mathcal{S}$ be the stability/firewall conditions from Appx_18 (e.g., suppression near forbidden zones)

Derive a confinement inequality of the form:

$$\sqrt{\det h(y)} \leq C \exp(-\Phi(y; \mathcal{B}, \mathcal{S}))$$

where $\Phi \rightarrow \infty$ as $|y| \rightarrow \infty$, with growth rate fixed by $\mathcal{B}, \mathcal{S}$.

If you can't express Φ in terms of upstream invariants, A2 is invalid.

3.5 Explicit Rejection of Tunable Warp Factors (Firewall)

Forbidden move:

- Choosing $f_a(y)$ with a free length scale L and then setting L “to make λ nice.”

That is post-fit behavior in disguise.

Kill switches (A2)

- `A2_FREE_DECAY_SCALE`

Trigger: any decay length / exponent introduced without upstream derivation.

- `A2_INTEGRABILITY_BY_TASTE`

Trigger: “assume $f \sim e^{-|y|/L}$ ” without a derivation of L .

- `A2_ONLY_RATIO_SAVED`

Trigger: A2 can only bound κ_7/λ by importing observational constraints.

Output of §3 (if invoked)

A short closure statement:

- Conditions on $h_{ab}(y)$ (or $\sqrt{\det h}$) that guarantee $\lambda < \infty$
- The exact list of kill switches above
- A clearly stated dependency: “A2 is valid only if decay is derived from

Appx_10/18 bounds, not chosen.”

§4 — GR Recovery Consistency Lock (B3 Arbiter)

Purpose

Determine whether κ_7 is an independent physical invariant or whether only the ratio κ_7/λ is meaningful once the reduction is done.

This section is a normalization lock: two independent routes to the 4D Einstein–Hilbert prefactor must agree. If they do not, the reduction channel is inconsistent and the gravity node cannot promote.

4.1 KK Reduction Normalization (7D $\rightarrow$ 4D)

Start from the 7D Einstein–Hilbert action written in the upstream normalization:

$$\begin{aligned} S_{\{7\}} &= \\ &\frac{1}{2\kappa_7} \int d^7x \sqrt{-g^{(7)}} R^{(7)} \\ &+ S_{\{\text{matter}\}}. \end{aligned}$$

Assume a product or weakly warped ansatz sufficient to define an effective reduction measure:

$$\begin{aligned} d^7x &= d^4x \, d^3y, \\ \sqrt{-g^{(7)}} &= \sqrt{-g^{(4)}} \sqrt{\det h} (\times \text{warp factor, if present}). \end{aligned}$$

Define the internal measure factor (as in §1):

$$\lambda := \int d^3y \sqrt{\det h} \, d^4x.$$

Under the standard KK truncation assumptions (no internal dependence of the retained 4D graviton mode), the leading 4D gravitational term reduces to:

$$\begin{aligned} S_{\{\text{KK}\}} &= \\ &\frac{\lambda}{2\kappa_7} \int d^4x \sqrt{-g^{(4)}} R^{(4)} + \cdots \end{aligned}$$

Match this to the canonical 4D Einstein–Hilbert form:

$$\begin{aligned} S_{\{4\}} &= \end{aligned}$$

$$\frac{1}{16\pi G_4} \int d^4x \sqrt{-g^{(4)}} R^{(4)}.$$

Therefore the KK normalization identity is:

$$\boxed{\frac{1}{16\pi G_4}} = \frac{\lambda^2 \kappa_7}{\quad} \quad \Longleftrightarrow \quad \boxed{G_4} = \frac{\kappa_7^2 \pi \lambda}{\quad}.$$

This is not yet a prediction; it is a bookkeeping identity contingent on the reduction assumptions.

4.2 Independent GR Recovery Channel (Appx_21)

Appx_21 asserts a separate recovery route: the effective long-wavelength dynamics in the external sector converge to 4D GR with a specific Einstein prefactor.

This channel can be stated as:

- There exists a regime (low curvature / long wavelength / regulator-consistent limit) in which the external equations of motion reduce to:

$$G_{\mu\nu}^{(4)} = 8\pi G_4 T_{\mu\nu}^{(4)}.$$

- The normalization of G_4 in that recovery is fixed by the same upstream conventions used to write the action and define stress-energy (i.e., no hidden rescalings post hoc).

Key point: Appx_21 supplies a normalization claim for the 4D Einstein equations without invoking the KK integral explicitly as the defining step. It is a recovery constraint, not a reduction identity.

4.3 Equality Constraint (Normalization Lock)

The consistency requirement is:

The G_4 obtained by the KK reduction normalization must equal the G_4 required by the GR recovery channel when both are expressed in the same upstream conventions.

Formally:

$$G_4^{\text{KK}} = G_4^{\text{Appx21}}.$$

Using §4.1:

$$G_4^{\text{KK}} = \frac{\kappa_7}{8\pi \lambda}.$$

So the lock condition is:

$$\boxed{G_4^{\text{Appx21}} = \frac{\kappa_7}{8\pi \lambda}}$$

Interpretation: This equation is not used to “solve for G_4 ” from measurement. It is used to check whether the same normalization emerges from two internal-consistency routes of the framework.

If the framework’s conventions make G_4^{Appx21} a fixed expression in upstream invariants, then the equality locks the admissible κ_7/λ .

If Appx_21 only fixes an equation form up to a multiplicative constant, then this lock exposes the missing normalization debt explicitly rather than hiding it.

4.4 Outcome Classification (What B3 Decides)

Case 1 — Ratio-Only Closure (Preferred)

If the recovery channel determines only the effective 4D prefactor, then the framework only ever needs:

$$\frac{\kappa_7}{\lambda} \quad \text{or equivalently} \quad \lambda/\kappa_7$$

and κ_7 is not independently physical within the 4D effective theory.

Operational consequence:

- κ_7 may be treated as a convention provided the convention is locked upstream and never tuned after evaluating λ .
- Prediction lives in the computed λ (and any other upstream invariants), not in sliding κ_7 .

This is the “ κ_7 collapses into a corollary” outcome.

Case 2 — Residual Normalization Debt (κ_7 remains independent)

If Appx_21 yields the GR form but does not fix the prefactor in terms of upstream invariants (or fixes it only up to an unfixed scale), then the equality constraint does not determine κ_7/λ uniquely.

Operational consequence:

- κ_7 remains an independent constant/modulus that must be fixed before λ evaluation.
- A separate derivation (B2-style) becomes mandatory only now.

This is not failure by itself — it is a declared debt. It becomes failure only if κ_7 is later fitted to force agreement with observed G .